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TECH HUB

• AI & PRODUCTIVITY • LESSON 09

AI Video Generation

Create Professional Videos with AI — No Camera Required

Lesson 09 of 30 | AI & Productivity Programme | AYE Tech Hub

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- How AI video generation works — diffusion models, transformers, and neural rendering
- Top 7 AI video tools compared: Runway, Pika, Sora, Kling, Hailuo, Luma, CapCut AI
- Text-to-video, image-to-video, and video-to-video workflows explained
- Prompt engineering for video — motion, style, duration, and camera movement cues
- AI avatars and talking heads — HeyGen, Synthesia, D-ID for presenter videos
- AI video editing tools — auto-captions, background removal, B-roll generation
- Practical use cases: YouTube, course content, marketing, explainer videos

What Is AI Video Generation?

AI video generation is the ability to create moving video content from text prompts, still images, or existing footage — without a camera, actors, or a video production team. In 2023, this technology was primitive and unreliable. By 2025, tools like **Runway Gen-3**, **Kling**, and **OpenAI Sora** can produce photorealistic, cinematic video clips indistinguishable from real footage. For content creators, educators, marketers, and engineers, this represents one of the most transformative capability shifts since the smartphone.

Unlike AI image generation (covered in Lesson 08), video generation must solve the additional challenge of **temporal consistency** — ensuring that objects, faces, lighting, and motion remain coherent across dozens or hundreds of frames. A face that looks perfect in frame 1 must look identical in frame 60. This is why video generation remained behind image generation for several years, and why it is still improving rapidly with each model release.

How AI Video Generation Works — The Technology

Most modern AI video generators use one of two core architectures, often combined:

Technology	How It Works	Strength	Used In
Diffusion Models (video)	Apply the same denoising diffusion process as image AI, but extended across time — each frame is generated while maintaining consistency with adjacent frames via temporal attention layers	Photorealistic quality; excellent for short clips; handles complex lighting	Runway Gen-3, Stable Video Diffusion, Pika
Transformer-based (video)	Treat video as a sequence of patches across space and time — similar to how text transformers process tokens. The model learns relationships between all parts of all frames simultaneously	Better long-range consistency; handles complex scenes; scales well with compute	OpenAI Sora, Kling (hybrid approach)
GAN-based (legacy)	Generator creates frames; discriminator judges if they look real. Faster than diffusion but lower quality and prone to artifacts	Fast generation; low compute requirement	Earlier tools (2020-2022), some real-time applications
Neural Radiance Fields (NeRF)	Learns a 3D representation of a scene from 2D images, then renders it from any viewpoint — creating smooth camera movement	Photorealistic camera fly-throughs; consistent 3D scenes	Luma AI, some Runway features, 3D background generation

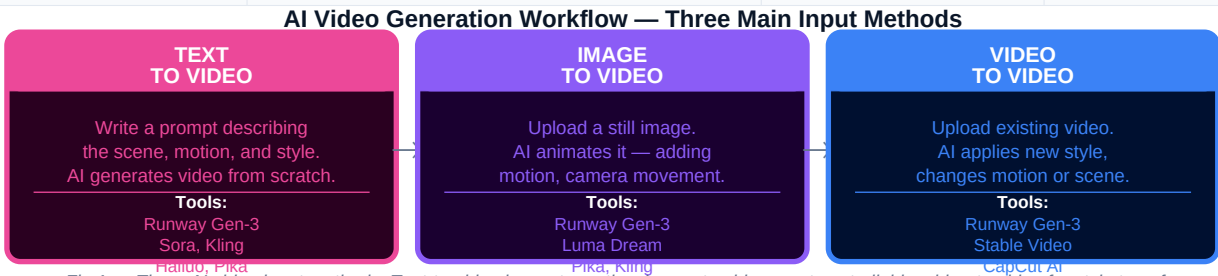
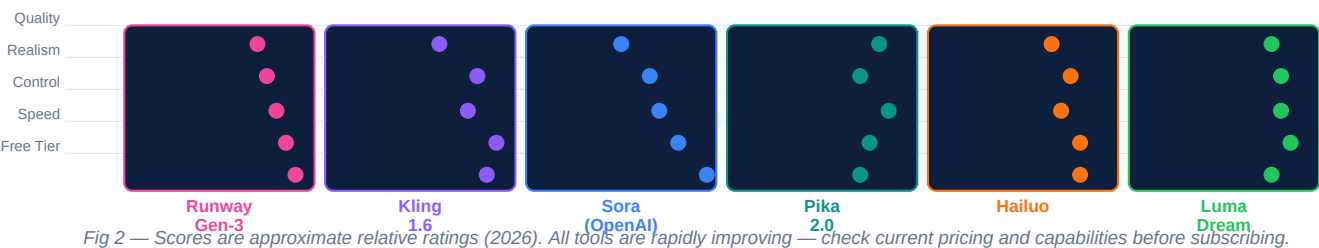


Fig 1 — Three AI video input methods. Text-to-video is most creative; image-to-video most controllable; video-to-video for style transfer

Top AI Video Generation Tools — 2025 Guide

The AI video tool landscape is moving extremely fast — new models are released every few months. The tools listed here were leading as of mid-2025. Always check current pricing and model versions before subscribing, as capabilities and costs change frequently.

Top AI Video Tools — Feature Comparison at a Glance



Tool	Best For	Max Duration	Resolution	Free Tier?	Approx Cost
Runway Gen-3 Alpha	Cinematic quality, professional content, VFX	10 sec	1280×768	Yes — 125 credits/month	\$12–\$76/month
Kling 1.6 (Kuaishou)	Realistic motion, longer clips, commercial use	2 minutes	1080p	Yes — limited daily	\$8–\$66/month
OpenAI Sora	Complex scenes, accurate physics, long duration	1 minute	1080p	Via ChatGPT Plus	Included in Plus (\$20)
Pika 2.0	Quick social media clips, fun effects, lip sync	5 sec	1080p	Yes — 150 videos/month	\$8–\$28/month
Hailuo (MiniMax)	Realistic human motion, consistent faces	6 sec	1080p	Yes — daily free credits	\$9–\$25/month
Luma Dream Machine	Smooth camera movement, scenic shots	9 sec (loop)	1080p	Yes — 30 videos/month	\$0–\$29.99/month
CapCut AI Video	Short-form social content, auto-editing, templates	15 sec	1080p	Yes — generous free tier	\$0–\$10/month

Which Tool Should You Start With?

Your Goal	Recommended Starting Tool	Why
Learning AI video — free first	Luma Dream Machine or Pika 2.0	Generous free tiers, easy interface, good quality for beginners
YouTube or course content	Runway Gen-3 or Kling	Best quality for longer professional clips; watermark-free on paid plans
Social media (Reels, TikTok)	CapCut AI + Pika	CapCut handles editing workflow; Pika generates clip inserts quickly

AI avatar presenter videos	HeyGen or Synthesia (see Section 4)	Dedicated avatar tools — Runway cannot generate consistent faces well yet
Marketing and ads	Kling + Runway combination	Kling for product shots and realism; Runway for cinematic quality and effects
Engineering explainers	Sora (via ChatGPT Plus) + CapCut	Sora understands technical scenes; CapCut adds captions and B-roll editing

Prompt Engineering for AI Video

The quality of AI-generated video is directly proportional to the quality of your prompt. A vague prompt produces generic, often incoherent output. A specific, structured prompt consistently produces professional-looking results. Video prompting has six key components — unlike image prompting, you must describe **motion** and **time** in addition to appearance.

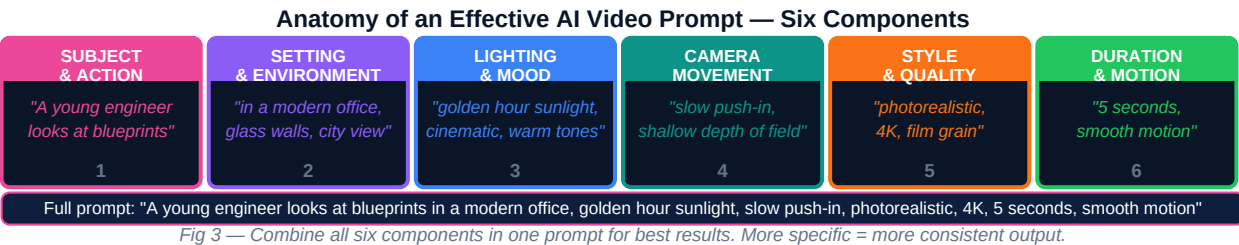


Fig 3 — Combine all six components in one prompt for best results. More specific = more consistent output.

Camera Movement Vocabulary — Essential for Video Prompts

Camera Term	What It Does	When to Use It	Example Prompt Phrase
Push in / Dolly in	Camera moves forward toward subject	Creating tension, intimacy, focus	"slow push-in toward the subject"
Pull out / Dolly out	Camera moves backward away from subject	Revealing context, showing scale, ending shots	"camera slowly pulls back to reveal the city"
Pan left / Pan right	Camera rotates horizontally on fixed axis	Following action, landscape reveals	"slow pan from left to right"
Tilt up / Tilt down	Camera rotates vertically on fixed axis	Revealing height of buildings, looking up at sky	"camera tilts up to reveal the skyscraper"
Orbit / Arc shot	Camera circles around the subject	Product reveal, dramatic subject introduction	"camera orbits slowly around the subject"
Aerial / Drone shot	High-angle top-down or sweeping overhead view	Establishing shots, landscape overview	"aerial drone shot sweeping over city"
Handheld / Shaky cam	Slightly unstable camera movement	Documentary feel, urgency, realism	"handheld documentary-style camera"

Static / Locked off	Camera completely fixed, no movement	Architectural shots, precise compositions	"static camera, no movement, locked frame"
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Style and Quality Keywords That Improve Output

Category	Keywords to Include	Effect on Output
Cinematic quality	"cinematic, 4K, film grain, anamorphic lens, shallow depth of field"	Output mimics professional film — bokeh background, letterbox feel, colour grading
Photorealistic	"photorealistic, ultra-realistic, hyperdetailed, natural lighting"	Maximises realism; avoids animated or stylised appearance
Lighting style	"golden hour, dramatic side lighting, soft diffused light, neon-lit night"	Controls mood and atmosphere — lighting is the most powerful visual cue
Time of day	"dawn, midday sun, dusk, night-time, blue hour"	Sets time context; affects colour temperature and shadow direction
Art direction	"minimalist, brutalist architecture, Blade Runner aesthetic, Studio Ghibli style"	Applies specific visual style — useful for stylised brand content
Motion quality	"smooth motion, fluid movement, no jitter, slow motion (120fps)"	Controls how the movement feels — essential for professional-looking clips
Avoid keywords	"blurry, distorted, watermark, low quality, extra fingers"	Negative prompts — use in Runway and other tools that support them

Prompt Improvement Examples — Before and After

Weak Prompt	Improved Prompt	Why It Is Better
"A man working"	"A focused engineer in his 30s reviews circuit diagrams at a sleek white desk, warm desk lamp, photorealistic, slow push-in, 4K, 5 seconds"	Subject defined (engineer), action specific (reviews diagrams), setting given, lighting named, camera move included, quality stated
"Show a city"	"Aerial drone shot slowly sweeping over the Addis Ababa skyline at golden hour, warm amber light, cinematic, smooth motion, high altitude perspective"	Shot type given (aerial drone), specific city, time of day, lighting, camera motion, quality — no ambiguity
"Product video"	"Close-up of a solar panel array on a rooftop, sun reflecting off cells, camera orbits slowly around the panels, photorealistic, bright midday sun, 8 seconds"	Product specified, environment given, camera orbit for drama, lighting precise, duration given — professional result

AI Avatars and Talking Head Videos

AI avatar tools solve a specific problem that general video generators cannot: creating a **consistent presenter** who delivers scripted speech on camera. Tools like **HeyGen**, **Synthesia**, and **D-ID** allow you to create a digital human presenter from a photo or pre-recorded video clip, then generate unlimited new videos of that presenter

speaking any script — in any language, with synchronized lip movement, in minutes.

Tool	What It Does	Key Advantage	Pricing	Best Use Case
HeyGen	Clone your own face and voice from a 2-minute video; generate talking videos from text scripts	Voice cloning included; your own likeness; 40+ languages; very natural lip sync	Free: 1 video/month; \$29–\$89/month paid	Course creators, YouTube educators, personalised sales videos, multilingual content
Synthesia	Library of 150+ professional AI avatars; text-to-speech with lip sync; screen recording overlay	No camera needed; enterprise-grade; SCORM export for LMS; branded templates	Free: 3 videos/month; \$22–\$67/month	Corporate training, e-learning, onboarding, explainer videos for businesses
D-ID	Animate any face photo to speak; natural expression and head movement from audio file	Works from a single photo; no video recording needed; API available	Free: 5 videos; \$5.90–\$49.90/month	Quick presenter videos from headshots, social content, chatbot avatars
Captions AI	Add AI avatars, auto-captions, eye contact correction, and filler word removal to real camera footage	Enhances real footage rather than replacing it; makes editing fast	Free tier; \$7.99–\$29.99/month	Content creators who film themselves; improving existing video quality quickly

How to Create a HeyGen AI Avatar Video — Step by Step

#	Step	Detail / Best Practice
1	Record your avatar video	Film yourself for 2–5 minutes, looking straight at camera. Good lighting, neutral background, no glasses if possible. Speak naturally — avoid reading from a script.
2	Upload to HeyGen and create avatar	Go to heygen.com → Avatar → Create Avatar. HeyGen processes your video in 30–60 minutes. Voice is automatically cloned.
3	Write your script	Type your script in the text box. HeyGen generates your avatar speaking it. Tip: write as you speak, not as you write — short sentences, natural pauses.
4	Choose background and layout	Select a background (built-in sets, or upload your own). Set avatar position — full body, half body, or circle overlay on a presentation.
5	Preview and adjust	Play the preview. Check lip sync, pacing, and pronunciation. Add pauses with [pause 1s] tags. Re-generate individual lines if needed.
6	Add slides or screen recording overlay	Upload PowerPoint slides. HeyGen overlays your avatar in corner while slides show. Alternatively, export video and add slides in CapCut or Canva.

7	Export and publish	Download MP4. Upload to YouTube, embed in course platform, or share directly. For multilingual: translate script → generate same avatar in new language.
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ETHICAL NOTICE — AI avatar technology can be misused to create deepfakes — videos of real people saying things they never said. Always disclose when presenting an AI-generated avatar in professional and public contexts. Use your own likeness or licensed avatars. Never create avatar videos of other people without their explicit consent. Most platforms now have detection systems for AI-generated content — violations can result in account termination and legal action.

AI-Powered Video Editing Tools

Beyond generating video from scratch, AI has transformed video editing — tasks that previously took hours of manual work in Premiere Pro or Final Cut can now be done in minutes. These AI editing tools work with your existing footage to enhance it, add elements, or restructure it automatically.

AI Editing Feature	Tool(s)	Time Saving	How to Use It
Auto captions and subtitles	CapCut, Descript, Veed.io, Adobe Premiere AI	80% faster than manual captioning	Upload video → AI transcribes speech → captions auto-synced. Edit transcript to fix errors. Export with captions burned in or as SRT file.
Background removal (AI matting)	CapCut, RunwayML, Adobe Express	Replaces green screen setup	Upload video with person in frame → AI removes background → replace with any image or video. Works without a green screen if lighting is reasonable.
Auto B-roll generation	Runway, Descript, OpusClip	Eliminates stock footage searching	AI watches your talking-head video → suggests and inserts relevant B-roll clips based on what you are saying. Review and approve each insertion.
Filler word removal	Descript, Gling, Adobe Premiere AI	Cuts editing time by 50%+	Upload video → AI detects "um", "uh", "like", long pauses → automatically removes them. Review remaining edits before export.
AI colour grading	DaVinci Resolve AI, Adobe Lumetri AI	Speeds up colour workflow 60%	AI analyses your footage → suggests or applies a colour grade matching a reference image or built-in style. Fine-tune manually after.
Clip repurposing	OpusClip, Vidy.ai, Gling	Creates 10 shorts from 1 long video	Upload a long video (webinar, interview, lecture) → AI identifies most engaging 60-second clips → auto-crops for vertical (Reels/TikTok) format with captions.
Voice enhancement and noise removal	Adobe Podcast AI, Descript Overdub, Krisp	Replaces need for professional recording studio	Upload audio or video with bad sound → AI removes background noise, echo, and hum. Adobe Podcast Enhance is free and works on a single file.

Practical Use Cases for Engineers and Content Creators

Use Case	AI Tools to Use	Workflow Summary	Time Saved vs Traditional
YouTube educational video	HeyGen + CapCut + Runway	Script → HeyGen avatar records → CapCut adds graphics and captions → Runway generates B-roll inserts	70% vs filming + studio editing
Online course content (LMS)	Synthesia + Canva + Veed.io	Slides in Canva → Synthesia adds avatar presenter → Veed adds captions → export SCORM	80% vs recording + post-production
Engineering explainer video	Sora + CapCut + D-ID	Sora generates technical animation → D-ID adds presenter avatar overlay → CapCut finalises and adds callouts	85% vs 3D animation + voice-over
Product or project proposal	Luma + Runway + CapCut	Luma generates aerial site flythrough → Runway creates interior visualisation → CapCut adds text overlays and narration	90% vs drone footage + animation studio
Marketing and social media	Kling + Pika + CapCut AI	Kling generates photorealistic product clips → Pika adds motion effects → CapCut auto-formats for Instagram/TikTok	75% vs videographer + video editor
Training and safety videos	Synthesia + Adobe Podcast AI + Veed	Synthesia avatar presents safety scenarios → Adobe Podcast AI cleans audio → Veed adds subtitles and compliance logo	80% vs actor + production team + studio

Recommended Starter Workflow — Create Your First AI Video in 30 Minutes

#	Step	Tool	Time
1	Write a 60-second script on your topic	ChatGPT / any text editor	5 min
2	Generate 3–4 background video clips from script keywords	Pika 2.0 (free)	5 min
3	Create an avatar reading your script	HeyGen (free trial)	5 min
4	Combine avatar + B-roll clips and add auto-captions	CapCut (free)	10 min
5	Add background music and text overlays	CapCut AI templates	3 min
6	Export and post (YouTube Shorts, Instagram Reels, or LinkedIn)	Platform of choice	2 min

IMPORTANT — AI video generation is improving so rapidly that the specific tool rankings in this lesson will change within months. The principles — prompting technique, workflow structure, and use case thinking — remain constant. Focus on mastering the workflow, not memorising which tool scores highest today. Check [ayetechub.com](https://www.ayetechub.com) for updated tool recommendations as AI video continues to evolve.

Key Takeaways — AI-009: AI Video Generation

- **AI video uses diffusion and transformer models** to maintain consistency across frames — this is why it was harder than image generation to perfect.
- **Three input modes:** text-to-video (creative), image-to-video (controlled), and video-to-video (style transfer). Mix them for best results.
- **Runway Gen-3, Kling, and Sora** lead for cinematic quality. Start free with Pika or Luma, then upgrade when you need longer or higher-quality clips.
- **Six-component video prompts** consistently outperform vague prompts: Subject, Setting, Lighting, Camera movement, Style/quality, Duration/motion.
- **AI avatars (HeyGen, Synthesia)** solve the presenter problem — create unlimited talking-head videos without filming yourself, in multiple languages.
- **AI editing tools save 70–90% of editing time** — auto-captions, background removal, filler word removal, and clip repurposing are ready to use today.
- **Next lesson — AI-010:** AI Voice and Avatar Tools — voice cloning, text-to-speech, AI podcasting, and audio-to-avatar workflows.